ROYAL RED CHILLI

DATA ANALYTICS PROJECT REPORT

Using MySQL and Power BI to understand sales, payments, and inventory.

■35,950	**■14,300**	**■21,950**
Total Sales	Gross Profit*	Payments Received
■14,000	**2,320 KG**	**■2,12,850**
Outstanding Amount*	Current Stock*	Current Stock Value*

Prepared as a portfolio project | Tools: MySQL, Power BI

*Values reflect the current project database and depend on its sample data, calculations, and relationships.

1. Executive Summary

Royal Red Chilli is represented in this analytics project as a business selling whole dry red chilli and chilli powder. The project combines relational data stored in MySQL with an interactive Power BI dashboard to help review sales, gross profit, payments, outstanding balances, and stock.

The dashboard presents a quick overview of the available data and includes monthly sales, product-wise sales, customer-wise sales, and filters for date, customer type, and product type. These views can help a business owner explore performance and identify areas that need attention.

2. Business Objectives

- Track total sales and estimated gross profit.
- Monitor payments received and outstanding balances.
- Estimate current stock quantity and stock value.
- Compare sales by product, customer, and month.
- Use structured data to support operational decisions.

3. Tools and Data Model

MySQL: stores the business data and supports SQL queries for aggregation, joins, and analysis.
Power BI: presents key performance indicators (KPIs), charts, and slicers in an interactive report.
Core tables: customers, products, suppliers, purchases, sales, sale_items, and payments.

4. Key Metrics

Metric	Value	Interpretation
Total Sales	■35,950	Sum of recorded sales amounts.
Gross Profit	■14,300	Sales line totals less product purchase-cost estimate.
Payments Received	■21,950	Sum of recorded payment amounts.
Outstanding Amount	■14,000	Total sales less recorded payments, assuming comparable records.
Current Stock	2,320 KG	Total purchases quantity less total sold quantity.
Current Stock Value	■2,12,850	Estimated value using the project's product purchase-price logic.

Important: validate metric definitions, relationships, and duplicate product variants before using these numbers for real business decisions. The values above are the current project figures, not independently audited financial results.

5. Dashboard Features

KPI cards: Total Sales, Gross Profit, Payments Received, Outstanding Amount, Current Stock, and Current Stock Value.

Monthly Sales: supports trend review across the dates available in the database.

Product-wise Sales: compares recorded sales across whole dry red chilli and chilli powder.

Customer-wise Sales: helps identify customers with higher recorded sales.

Interactive slicers: filter the report by date, customer type (such as retailer, wholesaler, or hotel), and product type.

6. Business Insights and Recommended Actions

• **Follow up on outstanding balances:** review unpaid sales by customer and payment date; confirm that payments are linked to the correct sales.
• **Monitor product performance:** compare quantities and sales value for each product and quality grade before deciding what to replenish.
• **Review stock valuation:** use a consistent costing method and calculate stock at the product-variant level.
• **Track monthly trends:** investigate unusually high or low sales periods and compare customer types.
• **Improve data quality:** ensure each sale item points to the correct product, purchase costs are current, and payment status matches actual collections.

7. Limitations and Data Checks

This is a portfolio analysis based on the current database. It should not be treated as audited financial reporting. The outstanding amount calculation assumes that recorded payments can be compared directly with recorded sales. The stock calculation assumes that purchase and sale quantities use consistent units and that returns, wastage, adjustments, and opening stock are not missing. Stock value should be checked against a defined inventory costing method and the product-variant relationships in the model.

8. Conclusion

The Royal Red Chilli project demonstrates an end-to-end analytics workflow: structuring business data in MySQL, querying it for useful measures, and presenting results through a Power BI dashboard. The next improvement is to document the SQL queries, include a dashboard screenshot, validate the data model, and publish the project files in a GitHub repository.

Portfolio note: Replace or supplement this report with a current screenshot of your actual Power BI dashboard before publishing.